Financial Analysis and Monitoring Agent

Financial markets generate a large amount of continuously changing information, including market data, corporate announcements, financial reports, and economic events. For investors and researchers, it is often difficult to continuously monitor relevant financial information, identify significant changes, and understand the potential reasons and implications behind them. This project proposes the development of a **Financial Analysis and Monitoring Agent** that can analyze financial data, monitor selected assets or companies, detect significant events, and autonomously investigate and explain their potential causes. The Agent should provide an interactive environment that supports financial analysis, monitoring, and decision-making.

Basic Requirements (70%)

- Financial Analysis
  - Retrieve financial and market data for selected stocks, funds, cryptocurrencies, companies, or other financial assets from external APIs or reliable data sources.
  - Calculate and analyze common financial indicators, such as returns, volatility, valuation ratios, profitability, and growth rates.
  - Support historical analysis and comparisons between different assets, companies, or time periods based on user-defined criteria.
- Financial Event Monitoring
  - Monitor selected financial assets, companies, or industries using external financial data sources.
  - Detect user-defined events or significant changes, such as large price movements, unusual trading volume, earnings announcements, or major corporate events.
  - When a significant event is detected, autonomously retrieve relevant information from appropriate data sources and analyze its possible causes and implications.
- Interactive Financial Analysis
  - Provide a user-friendly interface for users to configure monitoring targets, analysis tasks, and alert conditions.
  - Support natural-language follow-up questions and allow users to modify analysis or monitoring conditions interactively.
  - Present analytical results using appropriate tables, charts, and structured explanations.
  - Allow users to inspect the key data and information supporting the Agent's analytical conclusions.

Advanced Requirements (30%)

- Automated Anomaly Detection
  - Automatically identify unusual patterns or significant changes in financial data without requiring users to explicitly define every detection rule.
- Multi-step Investigation
  - When a significant event or anomaly is detected, autonomously decompose the investigation into multiple subtasks and dynamically determine what additional information or analysis is needed.
- Cross-asset Analysis
  - Analyze relationships between multiple financial assets, companies, or industries, such as relative performance, correlations, or common market drivers.
- Scenario Analysis
  - Allow users to modify selected assumptions or market conditions and analyze their potential impact on financial assets or portfolios.